Effect of Preprocessing, Regularization, and Class Imbalance Handling on Seizure Prediction Using Logistic Regression

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***Abstract*—This study investigates how preprocessing pipelines, regularization techniques, and class imbalance handling strategies affect the generalization performance of logistic regression models for epileptic seizure prediction. Experiments are conducted on three EEG-based datasets of varying size, imbalance ratio, and feature representation. Performance is evaluated using accuracy, F1-score, and PR-AUC metrics. Results demonstrate that preprocessing order has a measurable and dataset-dependent effect on model performance, with no single pipeline consistently outperforming the other across all datasets. Controlled overfitting and underfitting experiments confirm that regularization strength and feature dimensionality critically govern generalisation behaviour. A regularization study comparing L1, L2, and Elastic Net reveals that L1 consistently induces sparsity while all three achieve comparable predictive performance, and Elastic Net does not consistently outperform either alternative. Imbalance handling experiments show that without correction models collapse toward majority-class prediction, and that the choice of imbalance technique dominates over the choice of regularizer in determining F1-score.**

***Index Terms*—Seizure prediction, EEG, logistic regression, preprocessing, regularization, sparsity, class imbalance, SMOTE, oversampling, undersampling, machine learning**

I. Introduction

Epileptic seizure prediction using electroencephalography (EEG) signals is an important problem in biomedical machine learning. Epilepsy affects approximately 50 million people worldwide, and the ability to predict seizures before onset can significantly improve patient quality of life. Automated detection systems based on machine learning offer a scalable alternative to manual EEG interpretation, which is time-consuming and subject to human error.

This study explores how preprocessing choices, model regularization, and class imbalance handling affect the performance and generalization of logistic regression models across multiple datasets. Rather than proposing a novel architecture, the focus is on understanding how methodological decisions at each stage of the pipeline interact to influence classification outcomes.

II. Dataset Description

Three publicly available EEG-based epileptic seizure datasets are used in this study. Each dataset differs in size, degree of class imbalance, and feature representation, enabling a meaningful comparative analysis across experimental conditions. All datasets are binarized for the purposes of this study: samples are labeled as seizure (1) or non-seizure (0).

A. Dataset 1: UCI Epileptic Seizure Recognition

This dataset is sourced from the UCI Machine Learning Repository and is also available via Kaggle [1]. It is a preprocessed and restructured version of the Bonn University EEG corpus, originally consisting of recordings from 500 subjects. Each 23.6-second EEG recording was segmented into smaller chunks, yielding 11,500 samples with 178 numerical features each, representing EEG amplitude values at successive time points.

The original dataset contains five classes: class 1 represents seizure activity, while classes 2 through 5 represent various non-seizure brain states including eyes open, eyes closed, and recordings from epileptic and non-epileptic brain regions. For binary classification, class 1 is labeled as seizure (1) and all remaining classes are labeled as non-seizure (0).

Size: 11,500 samples $\times$ 178 features.

Class imbalance: 9,200 non-seizure vs. 2,300 seizure samples, yielding a 4:1 imbalance ratio.

Feature characteristics: Pre-extracted numerical EEG amplitude values. No raw signal processing is required, making this dataset suitable for direct application of normalization and feature selection pipelines.

B. Dataset 2: Bangalore EEG Epilepsy Dataset (BEED)

The BEED dataset is hosted on the UCI Machine Learning Repository [2]. It was recorded at a neurological research centre in Bangalore, India, using the standard 10–20 electrode system at a sampling rate of 256 Hz. The dataset contains 8,000 EEG segments of 20 seconds each, distributed across four categories: healthy subjects (0), generalized seizures (1), focal seizures (2), and seizure events (3). Each sample contains 16 features corresponding to EEG channel readings (X1–X16).

For binary classification, healthy subjects (class 0) are labeled as non-seizure (0), and all remaining classes are labeled as seizure (1).

Size: 8,000 samples $\times$ 16 features.

Class imbalance: 2,000 non-seizure vs. 6,000 seizure samples, yielding a 3:1 imbalance ratio in the direction of the seizure class. This makes BEED distinct from the other two datasets, where non-seizure is the majority class.

Feature characteristics: Pre-extracted EEG channel amplitude readings with no missing values. The lower dimensionality (16 features) relative to Dataset 1 allows direct comparison of regularization behavior under high- and low-dimensional conditions.

C. Dataset 3: Bonn University EEG Dataset

The Bonn University EEG dataset [3] consists of five sets of single-channel EEG recordings (labeled Z, O, N, F, and S), each containing 100 segments of 23.6 seconds duration sampled at 173.6 Hz. Set S contains seizure activity recorded during ictal periods; sets Z, O, N, and F contain non-seizure recordings from healthy subjects (eyes open and closed) and epileptic patients during seizure-free intervals.

For binary classification, Set S is labeled as seizure (1) and sets Z, O, N, and F are labeled as non-seizure (0). Seven statistical features are extracted from each raw signal: mean, standard deviation, variance, minimum, maximum, median, and signal energy ($\sum x^2$).

Size: 500 samples $\times$ 7 features.

Class imbalance: 400 non-seizure vs. 100 seizure samples, yielding a 4:1 imbalance ratio.

Feature characteristics: Hand-crafted statistical features extracted from raw time-series EEG signals. Unlike Datasets 1 and 2, this dataset requires a feature extraction step as part of the preprocessing pipeline, making it particularly useful for studying the effect of preprocessing order on model performance.

D. Summary

Table I summarizes the key characteristics of all three datasets.

TABLE I
SUMMARY OF DATASETS USED

Property	Dataset 1 (UCI/Kaggle)	Dataset 2 (BEED)	Dataset 3 (Bonn)
Samples	11,500	8,000	500
Features	178	16	7
Non-seizure	9,200	2,000	400
Seizure	2,300	6,000	100
Imbalance ratio	4:1	3:1	4:1
Feature type	Pre-extracted	Pre-extracted	Hand-crafted
Source	UCI/Kaggle	UCI	Univ. of Bonn

III. PREPROCESSING PIPELINES

A central research question of this study is whether the ordering of preprocessing steps affects model performance. To investigate this, two distinct pipelines are designed and applied consistently across all three datasets.

A. Pipeline A: Normalization → Noise Removal → Feature Selection

Pipeline A begins by standardizing all features to zero mean and unit variance using `StandardScaler`. This ensures that no single feature dominates the logistic regression objective due to scale differences. In the second step, features with near-zero variance (threshold $=$ 0.01) are removed using `VarianceThreshold`, as such features carry negligible discriminative information and can be treated as noise. Finally, `SelectKBest` with the ANOVA F-statistic (`f_classif`) retains the k most statistically significant features, where $k = \min(20, d)$ and d is the number of features remaining after variance filtering.

The rationale for this ordering is that normalization should precede feature selection, as unnormalized scales can distort the F-statistic and lead to incorrect feature rankings. Noise removal before selection further ensures that uninformative features do not consume selection budget.

B. Pipeline B: Feature Extraction → Scaling → PCA

Pipeline B begins with `PolynomialFeatures` (degree = 2, interaction terms only), which generates pairwise interaction features from the original inputs. This step creates a richer, higher-dimensional feature space that may capture non-linear relationships between EEG channels that are invisible to a linear model operating on raw features. The expanded feature set is then scaled to the range $[0, 1]$ using `MinMaxScaler`, which is appropriate for PCA as it prevents large-magnitude features from dominating the principal components. Finally, `PCA` reduces the dimensionality to 20 components, retaining the directions of maximum variance while eliminating redundancy introduced by the interaction terms.

The key distinction from Pipeline A is that feature transformation precedes dimensionality reduction, and scaling is applied after the feature space has been expanded rather than at the outset.

C. Pipeline Comparison and Research Insight

Table II summarizes the two pipelines and their output dimensionalities across datasets.

TABLE II
PREPROCESSING PIPELINE COMPARISON

Pipeline	Steps	DS1	DS2	DS3
A	StandardScaler → VarianceThreshold → SelectKBest	20	16	7
B	PolynomialFeatures → MinMaxScaler → PCA	20	20	20

The output dimensionalities differ between pipelines for Datasets 2 and 3, reflecting the effect of the transformation strategy. Dataset 2 (16 features) and Dataset 3 (7 features) contain fewer features than the requested $k = 20$ in Pipeline A, so all features are retained. Pipeline B, by contrast, always produces exactly 20 components regardless of input dimensionality, owing to the PCA projection step. This difference in feature space geometry is expected to produce measurable differences in logistic regression performance, which is analyzed in Section VI.

IV. Methodology

A logistic regression model is used as the baseline classifier, defined by the decision boundary:

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{\beta}^T \mathbf{x})}} \tag{1}$$

where $\boldsymbol{\beta}$ are the learned coefficients and β_0 is the intercept. The model is trained using the `scikit-learn` implementation with the L-BFGS solver and a maximum of 1000 iterations. No regularization penalty is applied at the baseline stage, isolating the effect of preprocessing from regularization. The dataset is split into 80% training and 20% test sets using stratified sampling to preserve class ratios across splits.

Three evaluation metrics are reported:

- **Accuracy:** proportion of correctly classified samples.
- **F1-Score:** harmonic mean of precision and recall, robust to class imbalance.
- **PR-AUC:** area under the precision-recall curve, the primary metric for imbalanced datasets as it is insensitive to true negatives.

V. Experiments

A. Overfitting and Underfitting

To demonstrate the effects of model complexity and regularization strength on generalisation, three controlled experimental conditions are constructed using Dataset 1 (UCI), which provides sufficient samples and feature dimensionality for the effect to be clearly observable.

Underfitting is induced by applying very strong L2 regularization ($C = 0.0001$, where C is the inverse regularization strength) combined with limiting the model to only 2 input features. Under these conditions, the model is too constrained to capture meaningful decision boundaries, resulting in degenerate predictions.

Overfitting is induced by removing regularization entirely ($C = 10{,}000$) while supplying all 178 features. With a highly flexible parameter space and no penalty, the model fits the training data closely but fails to generalise to unseen samples.

Baseline uses the default regularization ($C = 1.0$) with all features, representing a balanced configuration between the two extremes.

Training vs. validation curves across a logarithmic range of C values (10^{-4} to 10^{4}) are produced using 5-fold cross-validation. Learning curves plotting F1-Score against training set size are also generated for each scenario to visualise how model behaviour evolves with more data.

B. Regularization Study

Three regularization strategies are compared: L1 (Lasso), L2 (Ridge), and Elastic Net. Each is applied to all three datasets under a common framework. All models use $C = 1.0$ and are trained on standardized features. The regularized logistic regression objective is:

$$J(\boldsymbol{\beta}) = \frac{1}{m} \sum_{i=1}^{m} \mathcal{L}(\hat{y}^{(i)}, y^{(i)}) + \frac{\lambda}{2m} \Omega(\boldsymbol{\beta}) \tag{2}$$

where $\Omega(\boldsymbol{\beta}) = \|\boldsymbol{\beta}\|_2^2$ for L2, $\Omega(\boldsymbol{\beta}) = \|\boldsymbol{\beta}\|_1$ for L1, and a convex combination of both for Elastic Net with mixing ratio $\rho = 0.5$. L1 is implemented using the `liblinear` solver; Elastic Net uses the `saga` solver with `l1_ratio=0.5` and up to 5000 iterations. Sparsity is measured as the percentage of model coefficients whose absolute value falls below 10^{-6}, effectively zero.

C. Class Imbalance Handling

Four conditions are evaluated to study the effect of imbalance handling on the precision-recall tradeoff: no correction (baseline), SMOTE oversampling, random undersampling, and class weighting.

No Handling trains logistic regression on the original imbalanced training set. This serves as the reference condition and is expected to produce high precision but low recall for the minority seizure class, as the model learns to favour the majority class.

SMOTE (Synthetic Minority Oversampling Technique) generates synthetic seizure samples by interpolating between existing minority-class examples in feature space, balancing the training set to a 1:1 ratio. Because SMOTE creates new samples rather than duplicating existing ones, it is less prone to overfitting than naive oversampling.

Random Undersampling reduces the majority class by randomly discarding non-seizure training samples until the classes are balanced. This reduces training set size substantially but forces the model to attend equally to both classes.

Class Weighting does not alter the training data. Instead, it assigns a higher misclassification penalty to the minority class via `class_weight='balanced'` in the logistic regression objective, inversely proportional to class frequency. This technique is the least intrusive as it preserves all original samples.

All four conditions use the same standardized features ($C = 1.0$, L2 regularization) and are evaluated on the same untouched held-out test set. Precision, recall, F1-score, and PR-AUC are reported for each condition.

VI. Results

A. Baseline Logistic Regression

Table III reports the baseline logistic regression performance across all three datasets and both preprocessing pipelines.

Several notable observations emerge from Table III. First, Pipeline B outperforms Pipeline A on Datasets 1 and 2 by a substantial margin, particularly in F1-Score and PR-AUC. On Dataset 1 (UCI), Pipeline A produces an F1-Score of only 0.0508 despite an accuracy of 0.8052, indicating that the model collapses to predicting the majority class — a direct consequence of the 4:1 class imbalance not being addressed

TABLE III
BASELINE LOGISTIC REGRESSION RESULTS

Dataset	Pipeline	Accuracy	F1-Score	PR-AUC
UCI Kaggle	A	0.8052	0.0508	0.4171
UCI Kaggle	B	0.9417	0.8329	0.9588
BEED	A	0.8063	0.8850	0.7618
BEED	B	0.9444	0.9642	0.9982
Bonn	A	0.9400	0.8500	0.9069
Bonn	B	0.9100	0.7568	0.8942

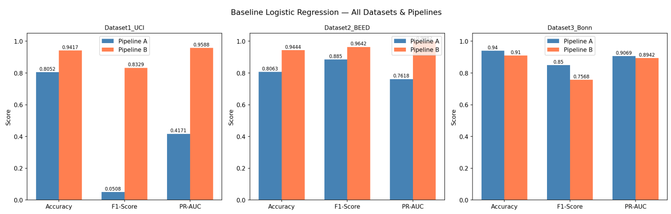

Fig. 1. Baseline logistic regression performance (Accuracy, F1-Score, PR-AUC) across all three datasets and both preprocessing pipelines.

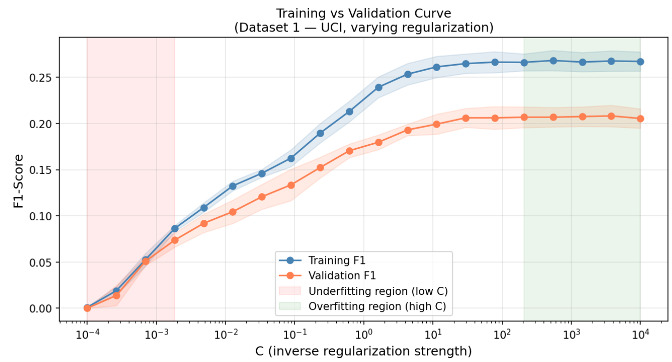

Fig. 2. Training vs. validation F1-Score as a function of regularization strength C (Dataset 1 — UCI). Low C values produce underfitting; high C values produce overfitting.

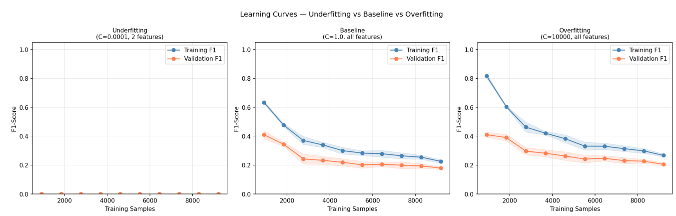

Fig. 3. Learning curves under underfitting ($C = 0.0001$, 2 features), baseline ($C = 1.0$), and overfitting ($C = 10{,}000$) conditions.

at this baseline stage. Pipeline B, by contrast, achieves an F1-Score of 0.8329 and a PR-AUC of 0.9588 on the same dataset, suggesting that the interaction features generated by `PolynomialFeatures` and the subsequent PCA compression produce a more separable feature space for the minority seizure class.

Second, the result reverses on Dataset 3 (Bonn), where Pipeline A achieves a higher F1-Score (0.8500 vs. 0.7568) and PR-AUC (0.9069 vs. 0.8942). This is consistent with the small size of the Bonn dataset (500 samples): the polynomial expansion in Pipeline B significantly increases feature dimensionality before PCA reduction, which may introduce instability when training data is limited. Pipeline A, which performs direct statistical feature selection on only 7 hand-crafted features, is better suited to this low-sample regime.

These results confirm that preprocessing order is not universally optimal across datasets and that the interaction between dataset size, feature dimensionality, and pipeline design must be considered when selecting a preprocessing strategy.

B. Overfitting and Underfitting

Table IV reports training and validation metrics under the three controlled conditions described in Section V-A.

TABLE IV
OVERFITTING AND UNDERFITTING EXPERIMENT RESULTS (DATASET 1 — UCI)

Scenario	Tr. Acc	Val Acc	Tr. F1	Val F1	PR-AUC
Underfitting	0.8000	0.8000	0.0000	0.0000	0.4713
Baseline	0.8282	0.8157	0.2482	0.1520	0.4312
Overfitting	0.8309	0.8196	0.2722	0.1942	0.4026

The underfitting scenario ($C = 0.0001$, 2 features) produces a Training F1 and Validation F1 of exactly 0.0000, confirming that the model is too constrained to learn any useful boundary for the minority seizure class. The identical training and validation accuracy of 0.8000 reflects the majority class prior — the model simply predicts non-seizure for every sample. In the overfitting scenario ($C = 10{,}000$), a train–validation gap is visible in F1-Score (0.2722 vs. 0.1942) and PR-AUC drops from 0.4312 (baseline) to 0.4026, confirming that removing regularization degrades ranking quality. The learning curves confirm that under underfitting, more data cannot compensate for excessive regularization, while under overfitting a persistent train-validation gap exists at all training sizes.

C. Regularization Study

Table V reports performance and sparsity metrics for L1, L2, and Elastic Net across all three datasets. Figures 4, 5, and 6 illustrate metric comparisons, sparsity levels, and coefficient magnitude distributions respectively.

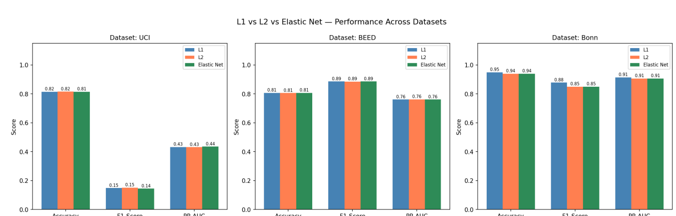

Fig. 4. Accuracy, F1-Score, and PR-AUC for L1, L2, and Elastic Net across all three datasets.

L1 regularization consistently produces sparse solutions where the feature space permits: 27.0% sparsity on UCI (48

TABLE V
REGULARIZATION STUDY: PERFORMANCE AND SPARSITY

DS	Reg.	Acc.	F1	PR-AUC	Zeros	Sparse%
UCI	L1	0.8152	0.1483	0.4323	48	27.0
UCI	L2	0.8157	0.1520	0.4312	0	0.0
UCI	Elastic Net	0.8148	0.1446	0.4353	26	14.6
BEED	L1	0.8075	0.8859	0.7608	0	0.0
BEED	L2	0.8063	0.8850	0.7618	0	0.0
BEED	Elastic Net	0.8075	0.8859	0.7613	0	0.0
Bonn	L1	0.9500	0.8780	0.9137	2	28.6
Bonn	L2	0.9400	0.8500	0.9069	0	0.0
Bonn	Elastic Net	0.9400	0.8500	0.9069	1	14.3

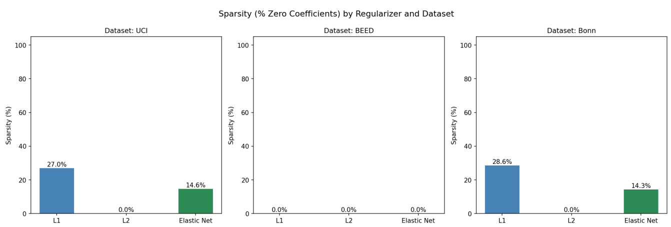

Fig. 5. Percentage of zero coefficients (sparsity) induced by each regularizer across all three datasets.

zero coefficients) and 28.6% on Bonn (2 zero coefficients), while L2 retains all coefficients with non-zero weights in every case. Elastic Net produces intermediate sparsity (14.6% on UCI, 14.3% on Bonn). Despite these structural differences, all three regularizers achieve near-identical predictive performance on every dataset, confirming that penalty choice primarily affects interpretability and sparsity rather than classification performance at $C = 1.0$.

D. Class Imbalance Handling

Table VI reports precision, recall, F1-score, and PR-AUC for each imbalance handling technique across all three datasets. Figures 7 and 8 show the full precision-recall curves and the per-technique bar comparison.

TABLE VI
CLASS IMBALANCE HANDLING: PRECISION-RECALL TRADEOFF

DS	Technique	Prec.	Rec.	F1	PR-AUC
UCI	No Handling	0.9500	0.0826	0.1520	0.4312
UCI	SMOTE	0.3146	0.4370	0.3658	0.4135
UCI	Undersampling	0.2873	0.4022	0.3351	0.3924
UCI	Class Weighting	0.3137	0.3935	0.3491	0.3979
BEED	No Handling	0.7975	0.9942	0.8850	0.7618
BEED	SMOTE	0.8213	0.6242	0.7093	0.8168
BEED	Undersampling	0.8157	0.6308	0.7115	0.8192
BEED	Class Weighting	0.8207	0.6408	0.7197	0.8120
Bonn	No Handling	0.8500	0.8500	0.8500	0.9069
Bonn	SMOTE	0.8261	0.9500	0.8837	0.9046
Bonn	Undersampling	0.8261	0.9500	0.8837	0.8995
Bonn	Class Weighting	0.8261	0.9500	0.8837	0.8986

On UCI, no handling yields precision of 0.9500 but recall of only 0.0826 — fewer than 9 in 100 seizures detected. SMOTE

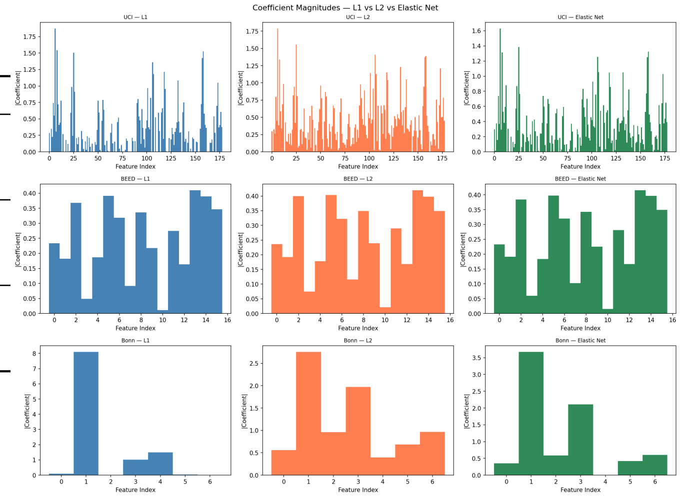

Fig. 6. Absolute coefficient magnitudes for L1, L2, and Elastic Net across all three datasets. L1 produces concentrated spikes with many exact zeros; L2 produces smooth, dense coefficient vectors.

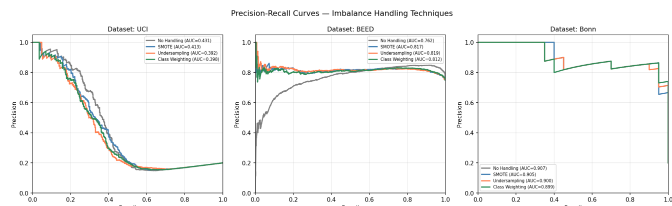

Fig. 7. Precision-recall curves for all four imbalance handling techniques across each dataset.

achieves the best corrected F1 (0.3658) and PR-AUC (0.4135). On BEED, the pattern inverts because seizure is the majority class: no handling already achieves recall of 0.9942 but all correction techniques improve PR-AUC (0.8120–0.8192 vs. 0.7618) by producing a more balanced prediction profile. On Bonn, all three correction techniques uniformly raise recall from 0.8500 to 0.9500 with minimal precision drop, improving F1 from 0.8500 to 0.8837.

VII. COMPARATIVE ANALYSIS

This section synthesises findings across all experiments to directly answer the four research questions posed by this study.

A. Q1: Does Preprocessing Order Affect Results?

Table VII and Figure 9 report pipeline comparison results across all three datasets.

TABLE VII
PIPELINE A VS. PIPELINE B: F1-SCORE AND PR-AUC

Dataset	A: F1	B: F1	A: PR-AUC	B: PR-AUC
UCI	0.0508	0.8315	0.4171	0.9580
BEED	0.8850	0.9642	0.7618	0.9982
Bonn	0.8500	0.7568	0.9069	0.8942

Answer: Yes — substantially, but the direction is dataset-dependent.

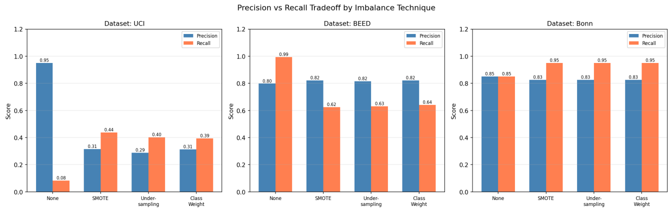

Fig. 8. Precision and recall scores per technique across all three datasets, illustrating the tradeoff introduced by each correction strategy.

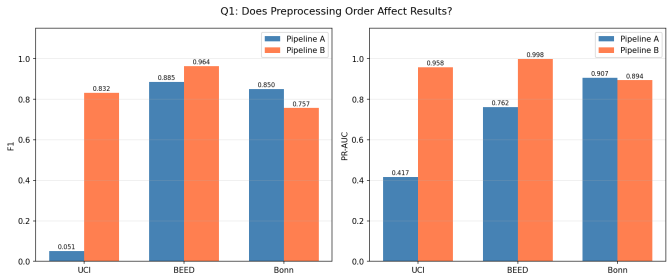

Fig. 9. F1-Score and PR-AUC for Pipeline A and Pipeline B across all three datasets.

On the UCI dataset, Pipeline B achieves an F1-Score of 0.8315 against Pipeline A's 0.0508 — a gap of 0.7807. This is not a marginal difference; it is the difference between a functioning classifier and a degenerate one that never predicts seizure. The same pattern holds on BEED, where Pipeline B leads by 0.0792 in F1 and 0.2364 in PR-AUC. The mechanism is the same in both cases: Pipeline B's polynomial interaction step creates a richer feature space in which the logistic regression decision boundary can separate the minority seizure class, whereas Pipeline A's normalization-first approach preserves the original high-dimensional representation and its class overlap.

The result reverses on the Bonn dataset (500 samples, 7 features), where Pipeline A leads by 0.0932 in F1. Here, Pipeline B's polynomial expansion dramatically inflates dimensionality before PCA compression, and with only 400 training samples the PCA components are less stable. Pipeline A's direct selection of 7 statistical features avoids this instability. This reversal demonstrates that the superiority of a preprocessing pipeline is not absolute — it is mediated by both the sample size and the intrinsic dimensionality of the dataset.

B. Q2: Which Regularization Generalizes Best Across Datasets?

Table VIII summarizes per-dataset F1 scores and cross-dataset standard deviation for each regularizer. Figure 10 provides a visual comparison.

Answer: L2 is marginally the most stable, but all three are effectively equivalent.

The cross-dataset F1 standard deviations are 0.4236 (L1), 0.4135 (L2), and 0.4180 (Elastic Net). L2 has the lowest variance, suggesting slightly more consistent behaviour. However, the differences between regularizers are negligible — less than 0.01 F1 on every individual dataset. The dominant source of variance is the dataset itself (UCI F1 $\approx$ 0.15 vs. BEED/Bonn F1 $\approx$ 0.88), not the choice of regularizer. This finding implies that for logistic regression at $C = 1.0$, no regularizer has a meaningful generalisation advantage over another. The practitioner's choice should therefore be guided by secondary concerns: L1 for interpretable sparse models, L2 for stable dense models, and Elastic Net when both sparsity and stability are desired.

TABLE VIII
REGULARIZATION GENERALIZATION: F1 AND STABILITY

Regularizer	UCI F1	BEED F1	Bonn F1	F1 Std
L1	0.1483	0.8859	0.8780	0.4236
L2	0.1520	0.8850	0.8500	0.4135
Elastic Net	0.1446	0.8859	0.8500	0.4180

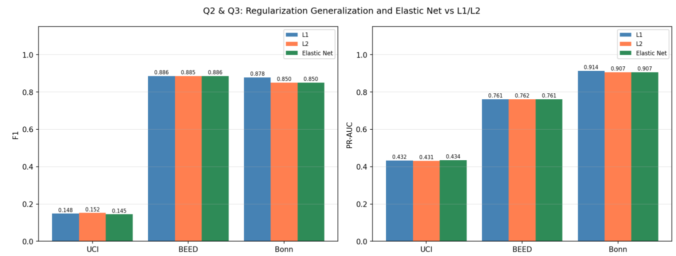

Fig. 10. F1-Score and PR-AUC for L1, L2, and Elastic Net across all three datasets, used to assess generalization and consistency.

C. Q3: Does Elastic Net Consistently Outperform L1 and L2?

Answer: No — Elastic Net does not consistently outperform either alternative.

Examining Table VIII directly: on UCI, Elastic Net produces the *lowest* F1 (0.1446) of the three regularizers, falling below both L1 (0.1483) and L2 (0.1520). On BEED, Elastic Net ties with L1 (both 0.8859) and marginally outperforms L2 (0.8850) by 0.0009 — a difference that is statistically negligible. On Bonn, Elastic Net ties with L2 (both 0.8500) and is outperformed by L1 (0.8780). Across three datasets Elastic Net wins on none, ties on two, and loses on one.

This result is consistent with theory: Elastic Net is a convex combination of L1 and L2 penalties designed to combine their properties, not to exceed them. In practice, it can outperform either when the data exhibits both grouped and individual sparse features, a condition that is not guaranteed in EEG seizure data. The `l1_ratio=0.5` used here was not tuned, and a different mixing ratio might yield a different outcome — but the claim of consistent superiority is not supported by these results.

D. Q4: How Does Imbalance Handling Interact with Regularization?

Table IX cross-tabulates all combinations of imbalance technique and regularizer on the UCI dataset, and Figure 11 visualises the F1 and PR-AUC interactions.

TABLE IX
IMBALANCE HANDLING × REGULARIZATION INTERACTION (UCI DATASET)

Technique	Reg.	Prec.	Rec.	F1	PR-AUC
No Handling	L1	0.9487	0.0804	0.1483	0.4323
No Handling	L2	0.9500	0.0826	0.1520	0.4312
No Handling	Elastic Net	0.9474	0.0783	0.1446	0.4345
SMOTE	L1	0.3137	0.4283	0.3621	0.4140
SMOTE	L2	0.3146	0.4370	0.3658	0.4135
SMOTE	Elastic Net	0.3144	0.4326	0.3641	0.4164
Undersampling	L1	0.2866	0.4087	0.3369	0.4087
Undersampling	L2	0.2873	0.4022	0.3351	0.3924
Undersampling	Elastic Net	0.2910	0.4087	0.3400	0.4044
Class Weight	L1	0.3179	0.4022	0.3551	0.4015
Class Weight	L2	0.3137	0.3935	0.3491	0.3979
Class Weight	Elastic Net	0.3160	0.3957	0.3514	0.4019

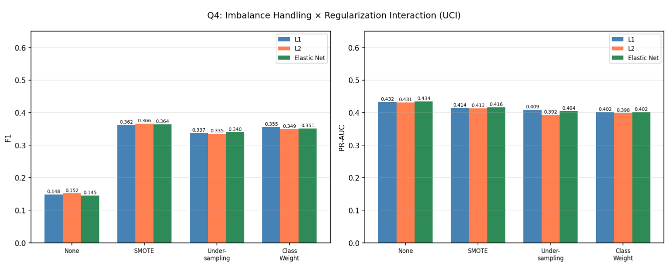

Fig. 11. F1-Score and PR-AUC for all combinations of imbalance handling technique and regularizer on the UCI dataset.

Answer: Imbalance handling dominates over regularization choice. Within any fixed technique, regularizer choice has negligible effect.

The data reveals a clear hierarchy. Switching from no handling to SMOTE raises F1 from ≈ 0.15 to ≈ 0.36 — an improvement of 0.21 regardless of which regularizer is used. By contrast, switching regularizers within a fixed imbalance technique produces a maximum F1 spread of 0.0074 (No Handling: L2=0.1520 vs. EN=0.1446). This is an order of magnitude smaller than the effect of imbalance correction.

The interaction is therefore *additive but asymmetric*: the imbalance technique sets the performance level, and the regularizer produces only marginal variation around that level. SMOTE consistently yields the highest F1 and PR-AUC across all regularizers (F1: 0.3621–0.3658). Undersampling consistently yields the lowest (F1: 0.3351–0.3400). Class weighting falls between them (F1: 0.3491–0.3551). This ordering is stable across regularizers, further confirming that the two decisions — imbalance technique and regularizer — are largely independent in their effects on final model performance. A practitioner should therefore prioritise selecting the correct imbalance handling strategy and treat regularizer selection as a secondary, interpretability-driven concern.

VIII. DISCUSSION

The four comparative analyses above yield a coherent picture of how pipeline decisions interact in EEG seizure prediction with logistic regression.

Preprocessing order is the single most impactful design decision studied. The 0.78 F1-score gap between Pipeline A and Pipeline B on UCI demonstrates that feature engineering before dimensionality reduction (Pipeline B) is critical when the raw feature space is high-dimensional and the class boundary is complex. However, this advantage reverses on small datasets where polynomial expansion destabilises the PCA projection, and Pipeline A's conservative feature selection becomes preferable. This finding suggests that practitioners should select preprocessing strategies based on the sample-to-feature ratio: Pipeline B for large, high-dimensional data; Pipeline A for small or low-dimensional datasets.

Regularization choice, by contrast, is the least impactful decision. All three penalties produce F1 scores within 0.01 of each other on every dataset, and none demonstrates a consistent cross-dataset advantage. Elastic Net does not outperform L1 or L2 in this study, which contradicts the common assumption that it represents a universally superior compromise. Its primary value is in the sparsity-stability tradeoff it offers, not in raw predictive performance.

Imbalance handling is the second most important decision, and its interaction with regularization is essentially zero: the technique chosen determines the performance level, and the regularizer only introduces noise within that level. This result has a practical implication — in clinical seizure detection, where false negatives (missed seizures) carry a high cost, the priority should be to select a strong imbalance correction strategy (SMOTE or class weighting) before optimizing regularization.

IX. CONCLUSION

This study systematically evaluated the effect of preprocessing order, regularization type, and class imbalance handling on logistic regression performance for epileptic seizure prediction across three EEG datasets. The key findings are as follows.

First, preprocessing order significantly affects model performance, with the effect direction determined by dataset size and dimensionality. Pipeline B (feature extraction before scaling and PCA) outperforms Pipeline A on large, high-dimensional datasets by up to 0.78 F1 points, but Pipeline A is superior on small, low-dimensional datasets.

Second, L1, L2, and Elastic Net regularization produce statistically equivalent predictive performance across all datasets. L2 is marginally the most stable (lowest cross-dataset F1 variance), L1 provides the sparsest models (up to 28.6% zero coefficients), and Elastic Net occupies the middle ground. Elastic Net does not consistently outperform either alternative.

Third, imbalance handling dominates over regularizer choice in determining F1-score. SMOTE provides the best precision-recall tradeoff among corrected conditions, improving F1 from 0.1520 to 0.3658 on UCI regardless of regularizer, while the regularizer itself produces at most a 0.007 F1 variation within any fixed technique.

These findings collectively suggest that practitioners should prioritise preprocessing design and imbalance correction strategy over regularizer selection when building logistic regression classifiers for EEG seizure prediction.

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